Part C: Experiments and Analysis

Experiment 1: Belady's Anomaly

Reference string: 1 2 3 4 1 2 5 1 2 3 4 5. Run once with three frames and again with four frames. Algorithm Faults (3 frames) Faults (4 frames)

1. FIFO: 9 10
2. LRU: 10 8
3. Optimal: 7 6

Question 1:

Does any algorithm produce more faults with more frames?

Answer:

Yes. FIFO increases from 9 faults with three frames to 10 faults with four frames in this experiment. This is called Belady's anomaly.

Question 2:

Why do LRU and Optimal not show this behaviour?

Answer:

LRU and Optimal satisfy the stack property under the standard model. At corresponding steps, the pages resident with fewer frames are a subset of the pages resident with more frames. Thus, increasing the number of frames cannot increase faults for these algorithms.

Experiment 2: Own Reference Strings

The assignment requires two reference strings of at least 15 references: one where LRU is better than FIFO, and one where FIFO is better than or equal to LRU. The candidate cases below use three frames; verify their counts by running the notebook.

Case Reference string

Frames FIFO faults LRU faults

LRU better

- 4 4 4 2 3 4 2 5
- 4 2 1 4 5 5 4
- 3 9 6

FIFO better/equal

- 1 4 3 5 4 1 3 5
- 4 3 1 2 2 3 5
- 3 9 11

Short Questions

Question:1

Why can Optimal not be implemented in a real OS?

Answer:

It requires exact knowledge of future page references, which a real operating system cannot generally know in advance.

Question:2

What extra bookkeeping does LRU need compared with FIFO?

Answer:

LRU tracks when each page was last used, using timestamps, counters or an ordered structure.

Question:3

FIFO mainly tracks arrival order. If frames equal distinct pages, how many faults occur?

Answer:

Assuming all frames start empty and frame count equals the number of distinct pages, every distinct page faults once on its first reference. No replacement is needed after all distinct pages have been loaded. Faults equal the number of distinct pages .